



# Analysis of the Impacts Between Infrastructure Performance and Logistics of Anggrek Port on Economic Growth in Gorontalo Province

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**Abstract:** As a feeder port for the Eastern Indonesia region, Anggrek Port is expected to reduce logistics frictions and stimulate regional growth in Gorontalo Province. This study examines how port infrastructure performance influences logistics performance and economic growth by employing a cross-sectional survey ( $n = 150$ ) involving managers, service providers, and users, analyzed using PLS-SEM (SmartPLS 4.0). The reflective measurement model meets conventional reliability and validity thresholds, and the structural relationships were assessed through bootstrapping. The findings indicate a strong and significant direct effect of port infrastructure performance on economic growth ( $\beta = 0.679$ ,  $p < 0.00$ ;  $R^2 = 0.574$ ), whereas no significant effects were identified between infrastructure and logistics performance ( $\beta = 0.236$ ,  $p = 0.256$ ) or between logistics performance and economic growth ( $\beta = 0.192$ ,  $p = 0.375$ ). These results underscore that the dynamics linking port infrastructure, logistics performance, and economic growth cannot be fully understood through a purely linear structural approach. Based on the observed relational patterns, the policy implications highlight that enhancements to Anggrek Port's physical infrastructure currently generate more immediate and substantial economic impacts than improvements to its logistical systems.

**Keywords:** Port infrastructure performance; Port logistics integration; Regional development; Gorontalo Province; Economic growth; PLS-SEM; Maritime transport policy

## 1 Introduction

Enhancing port infrastructure is crucial for contemporary maritime transportation systems, as it significantly influences logistics efficiency and the economic competitiveness of a region. Ports do not merely serve as cargo handling points but function as strategic nodes within global supply chain networks, connecting the hinterland to international markets [1]. Globally, high-performing ports such as Shanghai, Singapore, and Rotterdam have become catalysts for economic growth due to their ability to reduce logistics costs and facilitate trade flows. In Indonesia, an archipelagic nation with more than 17,000 islands, the role of ports is indispensable for ensuring inter-island connectivity, improving national logistics efficiency, and supporting integration into the global economy. However, the 2023 Container Port Performance Index (CPPI) [2] ranks Indonesia's major ports, such as Tanjung Priok (281) and Belawan (214), far below regional benchmark ports like Tanjung Pelepas (6) and Singapore (18), highlighting substantial gaps in productivity and efficiency.

Empirical evidence from various countries has demonstrated a positive association between port infrastructure efficiency and economic growth [3, 4]. Nevertheless, most Indonesian ports continue to perform suboptimally. According to the same report [2], ports such as Tanjung Perak (ranked 97), Belawan (ranked 214), and Tanjung Priok (ranked 281) lag significantly behind regional competitors, including Singapore (18) and Malaysia's Port of Tanjung Pelepas (6). Additionally, Indonesia's position in the Logistics Performance Index (LPI) remains relatively weak, with a score of 3.0 in 2023, placing the country sixth in the ASEAN region behind Singapore (4.3), Malaysia (3.6), Thailand (3.5), the Philippines (3.3), and Vietnam (3.3). The World Bank's LPI ranks Indonesia 63rd out of 160 countries, indicating that despite the substantial potential of its logistics sector, the quality of infrastructure and logistics efficiency still require improvement to support optimal future economic growth.

Within the national port hierarchy established in the National Port Master Plan (KP 432/2017) [5], port roles are differentiated into main ports, collector ports, and feeder ports. Anggrek Port in Gorontalo Province is classified as

a collector (feeder) port supporting major ports such as Bitung and Makassar and serving as a connector between the North Gorontalo hinterland and domestic as well as international trade routes. The Anggrek Port Master Plan (KM 87/2022) [6] further positions the port as a supporting facility for the Gopandang Special Economic Zone (SEZ). Despite this designation, the port continues to face limitations in cargo-handling facilities, operational efficiency, and intermodal integration, indicating that its full potential has not yet been realized.

Cross-country studies have consistently demonstrated linkages between port infrastructure efficiency and economic growth [7–11]. However, many Indonesian ports still occupy suboptimal positions in global comparisons [2]. The feeder system in Eastern Indonesia, including Gorontalo, plays a strategic role within the national supply chain but continues to suffer from underdeveloped infrastructure capacity and lagging logistics efficiency. These conditions create an analytically distinct context compared with major hub ports.

The geographical characteristics of Gorontalo further reinforce the relevance of this study. As a province with limited maritime gateways and a high dependence on sea transport, the performance of Anggrek Port directly affects logistics distribution, commodity prices, and regional economic stability. Moreover, Gorontalo exemplifies the broader characteristics of feeder ports in Eastern Indonesia: strategically important yet constrained by limited infrastructure and relatively inefficient logistics systems compared with western regions. Thus, the case of Anggrek Port offers an empirical proxy for understanding the dynamics of mid-level feeder ports in Eastern Indonesia.

Although a growing body of research has examined major hub ports (e.g., Tanjung Priok, Bitung, Surabaya), empirical studies on feeder or collector ports, particularly regarding how infrastructure performance influences logistics performance and ultimately regional economic growth, remain scarce [12–14]. This gap is conceptually significant because governance structures, scale, and network functions in feeder ports differ markedly from those of hub ports. These distinctions may alter the relational patterns among infrastructure, logistics, and economic growth as proposed in existing literature [7–9]. Accordingly, this study positions feeder ports as the analytical unit to address this research gap.

Anggrek Port was selected not only because of its role as Gorontalo's primary maritime gateway but also due to its formal designation within the national port governance system (KP 432/2017) [5] and its clearly defined development mandate (KM 87/2022) [6]. The port faces operational and intermodal challenges typical of feeder systems, while Gorontalo's economy is highly sensitive to port performance. These conditions strengthen the identification of both direct and indirect effects predicted by theory between infrastructure, logistics, and economic growth. Therefore, Anggrek Port provides an analytical context suitable for generalizing feeder-port dynamics beyond geographical considerations alone.

Based on the aforementioned rationale and research gap, this study formulates four central research questions:

- (1) To what extent does port infrastructure performance in feeder ports affect regional economic growth?
- (2) Does port infrastructure performance significantly enhance logistics performance in the feeder-port context?
- (3) Does logistics performance mediate the relationship between port infrastructure performance and regional economic growth?
- (4) What factors influence the relationship between port infrastructure performance, logistics performance, and economic growth?

The scope of this study is limited as follows:

- (1) The research is conducted solely at Anggrek Port (one port);
- (2) The study examines indicators at a general level;
- (3) The assessment of economic growth considers only port-related impacts, excluding other external factors.

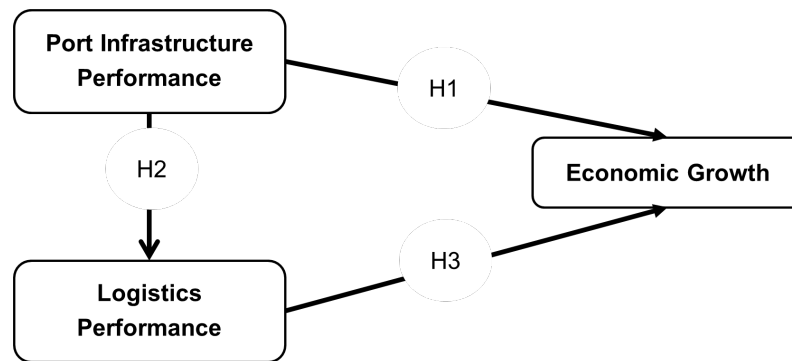
The primary innovation of this study lies in its methodological approach, which integrates empirical analysis using Structural Equation Modeling (SEM) to map causal relationships among port infrastructure performance, logistics performance, and regional economic development. This approach provides a conceptual contribution to transport economics and regional development studies, while also offering an evidence-based policy foundation for local governments to strengthen the strategic role of Anggrek Port within the national port network.

## 2 Conceptual Framework

This study seeks to explore how the infrastructure performance of Anggrek Port influences the economy of Gorontalo Province, particularly in achieving optimal logistics performance. Acknowledging the substantial disparity between the port's current conditions and its potential, the study posits that improvements in port infrastructure can also enhance the performance of the logistics sector. High-quality port infrastructure and logistics systems can strengthen accessibility to both local and global markets. Effective utilization of these market opportunities is expected to increase interprovincial and international trade, which in turn stimulates economic growth in the region [7]. A brief illustration of this relationship is presented in Figure 1.

First, this study examines the direct effect of port infrastructure performance on economic growth. It then evaluates the indirect effect of port infrastructure performance on logistics performance as the primary mediating

mechanism influencing economic growth. Based on Figure 1, three hypotheses are formulated to represent each of these relationships.



**Figure 1.** Hypothesized conceptual framework

### 3 Hypothesis Development

Consistently providing high-quality port infrastructure is regarded as an essential indicator of logistics performance and port management success. The core elements of port infrastructure include berth length, the number of available gantry cranes, vessel berthing facilities, port logistics information systems, terminal areas, and storage warehouses [15]. According to Pradhan and Bagchi [8], improvements in transportation infrastructure can stimulate economic growth by increasing aggregate demand. In addition, Ma et al. [16] found that port consolidation and integration statistically promote economic growth in port cities, particularly in small and medium-sized cities. Evidence from Karimah and Yudhistira [17] shows that the opening of small ports increases night-light intensity, a proxy for local economic activity, by 1.8%, providing empirical confirmation that port investments can stimulate local economic activities. Meanwhile, Jung [18] and Deng et al. [19] argue that ports exert trickle-down effects on regional economies. Similarly, Mudronja et al. [20] indicate that port throughput (cargo/container traffic) contributes positively to GDP or regional output and generates spillover effects into surrounding areas. Investments in transportation infrastructure offer benefits that extend beyond improving travel-time efficiency [21]. As highlighted by Lakshmanan [7], enhancing the quality of freight transport services can stimulate trade expansion, which subsequently increases labor supply and facilitates technology diffusion. Several studies on the economic impacts of port activities [22–26] demonstrate that port operations significantly influence both regional and national economies. For instance, Bottasso et al. [22] examined how port activity affects employment levels in 560 regions across 10 Western European countries and found that a one-million-ton increase in port throughput can generate approximately 400 to 600 local jobs. According to Bottasso et al. [23], a 10% increase in port throughput has the potential to raise regional GDP by 6–20%, while also promoting economic growth in neighboring regions by 5–18%. In China, Shan et al. [25] reported that a 1% increase in port cargo throughput can result in a 7.6% rise in per capita GDP, along with positive spillovers to adjacent regions. Similar findings were reported by Chang et al. [26], who indicated that the South African economy could experience a 17% decline in the absence of one unit of port activity. Furthermore, Park and Seo [27] found that container throughput volume in South Korean ports has a positive impact on regional economic growth. Therefore, the following hypothesis can be proposed:

**Hypothesis 1.** Port Infrastructure Performance Positively Influences Economic Growth.

According to Lakshmanan [7], investment in transport infrastructure can strengthen logistics capability and reduce the costs associated with the movement of goods. Meanwhile, Wilmsmeier and Hoffmann [28] explored the effects of Liner Shipping Connectivity (LSC) and port infrastructure on freight rates in the Caribbean region. Their findings indicate that a one standard deviation increase in LSC can reduce shipping costs by approximately USD 287. Similarly, an equivalent improvement in the quality of port infrastructure in the importing country can lower freight charges by about USD 225. Furthermore, Sánchez et al. [29] found that ports with higher levels of efficiency tend to have lower shipping costs after accounting for factors such as distance, the availability of shipping services, commodity type, and insurance costs. According to Clark et al. [1], increasing port efficiency from the 25th to the 75th percentile can reduce shipping costs by up to 12 percent. Infrastructure quality and transportation costs also play a critical role in fostering export-driven economic growth [30]. Consequently, efficiently managed ports typically possess superior infrastructure and logistics capabilities compared with less efficient ports. Efficient port systems with strong logistical capacity are also a key factor in attracting foreign direct investment [31]. Conversely, inefficient ports can hinder domestic and international trade activities and negatively affect economic growth [1]. In addition, Ellis [32] emphasized the crucial role of ports in supporting firm internationalization. Accordingly, the

following hypothesis is proposed:

**Hypothesis 2.** Port Infrastructure Performance Positively Affects Logistics Performance.

According to Munim and Schramm [9], national logistics efficiency plays a crucial role in streamlining the delivery of goods to global markets. However, when logistics services fail to operate optimally, trade activities are hindered by increased time requirements and financial costs [33]. Operational efficiency at ports stimulates investment flows into terminal operations, warehousing, and industrial zones in the hinterland, while also serving as a catalyst for the emergence of manufacturing clusters and logistics service providers. The subsequent impact of this process includes enhancements in regional productivity and job creation [20]. According to Limão and Venables [30], increases in transportation costs exhibit a significant negative correlation with international trade. Delays in customs procedures have been shown to reduce firms' total factor productivity [34]. A more conducive business environment, particularly when supported by a high-quality logistics system, is associated with improved national export performance [35]. Enhanced transport accessibility expands employment opportunities in the logistics sector [36]. Overall, the presence of a more effective national logistics system encourages firms to engage more proactively in exporting to international markets and enhances a country's attractiveness to foreign investors [37]. In conclusion, improvements and advancements in the logistics sector can substantially foster regional economic growth [10, 11, 38, 39]. Based on the findings of Millán et al. [40], a 1% increase in the LPI is associated with an increase in global economic growth ranging from 1.1% to 3.4%. Accordingly, the following hypothesis is proposed:

**Hypothesis 3.** Logistics Performance Positively Affects Economic Growth.

## 4 Methods

### 4.1 Population and Sample

The population of this study comprises three main stakeholder groups actively involved in Anggrek Port: port management, consisting of KSOP Anggrek and PT Anggrek International Terminal (AGIT); port service providers, namely shipping agency companies; and service users, including freight forwarding companies (JPT) and cargo owners.

The list of companies and institutions registered as active under KSOP Anggrek and PT AGIT during the year of study served as the basis for developing the sampling frame. The total population is estimated at approximately 200 individuals directly involved in port operations. From this population, 150 respondents were selected through purposive sampling, based on the consideration that they possess the most relevant knowledge regarding the port system and its operational activities. The sample comprised 50 participants from the management category, 50 from service providers, and 50 from service users. This sample size aligns with the recommended minimum thresholds for PLS-SEM, which suggest between 30 and 300 observations for analyses involving relatively simple model structures [41, 42].

### 4.2 Data Collection

Primary data were collected through questionnaires administered directly at Anggrek Port over a three-week period. Respondents were first briefed on the purpose of the study and assured of the confidentiality of their responses, which encouraged them to provide open and reliable information. Secondary data were obtained from several official sources, including KSOP Anggrek archives, PT AGIT documentation, and port statistical records, to support the empirical analysis.

This study employed a structured questionnaire adapted from previous research on port operations and regional logistics performance [9, 23, 29, 30, 43]. It consisted of 33 statements distributed across three core constructs: port infrastructure performance, logistics performance, and their influence on regional economic growth. Responses were measured using a seven-point semantic differential scale, ranging from 1 (very poor) to 7 (very good). The semantic differential scale is a bipolar measurement approach that captures connotations or perceptions of a concept by positioning pairs of antonyms (e.g., slow-fast, unreliable-reliable, poor-good) at opposite ends. According to Chen et al. [44], semantic differential scales often provide stronger internal reliability and validity than Likert scales when used to assess service quality, making them suitable when the objective is to capture bipolar nuances in respondents' perceptions. Therefore, this method was selected for data collection.

### 4.3 Data Analysis

Data were analyzed using SEM with SmartPLS 4.0, a method selected for its capability to evaluate causal relationships among latent variables within both reflective and formative model structures. In this study, a reflective measurement model was employed. Within reflective constructs, indicators function as manifestations of the latent variable, meaning that changes in the latent construct are proportionally reflected across all associated indicators [45]. Because indicators represent outcomes of the construct, they tend to exhibit strong intercorrelations and are often interchangeable. Reflective models are generally applied to measure intangible concepts such as perceptions,

satisfaction levels, and loyalty [46]. The evaluation of reflective constructs focuses on assessing the validity and reliability of the measurement instruments, encompassing the following procedures:

(1) Convergent Validity Test

Convergent validity was assessed by examining both outer loadings and the average variance extracted (AVE). Indicators are considered valid when their loading values reach at least 0.708, and their AVE values are 0.50 or higher [47]. For indicators with loadings ranging between 0.40 and 0.708, Chin [41] and Hair and Alamer [48] recommend that they should not be removed automatically. Instead, researchers should consider whether these indicators remain theoretically meaningful and empirically useful in representing the latent variable.

(2) Internal Consistency Reliability

Composite reliability (CR) and Cronbach's alpha were used to assess internal consistency reliability. Values greater than 0.70 are recommended to indicate that the indicators consistently capture the underlying latent construct [48].

(3) Discriminant Validity Test

Discriminant validity was evaluated using the Fornell–Larcker criterion and the Heterotrait–Monotrait Ratio (HTMT). A construct is considered to have satisfactory discriminant validity when the HTMT value is below 0.85, indicating that it can be empirically distinguished from other constructs [49].

(4) Structural Model (InnerModel)

Once the measurement model was confirmed valid and reliable, the next step involved evaluating the structural model through collinearity assessment (VIF), path coefficients,  $R^2$ , and  $f^2$  values [48].

## 5 Results

### 5.1 Measurement Model (Outer Model)

The first stage of the measurement model analysis involved assessing convergent validity and evaluating internal consistency reliability, which were examined through the factor loading values. The results of these assessments are summarized in Table 1.

VIF results indicate that all indicators fall below the practical threshold, suggesting that no multicollinearity is present within the constructs. In addition, Table 1 shows that all factor loading values meet the recommended threshold proposed by Hair and Hult [47], which states that indicators are considered valid when their loadings are  $\geq 0.708$ . This confirms that each indicator sufficiently represents its corresponding construct and satisfies the requirements for convergent validity. The AVE values for all constructs exceed 0.50 [48], indicating that the latent variables explain the majority of variance in their indicators. Accordingly, the measurement model can be considered to have achieved convergent validity. Therefore, it can be concluded that each construct demonstrates satisfactory convergent validity, with its indicators consistently capturing the underlying conceptual dimensions.

The next stage of the measurement model evaluation focuses on internal reliability, assessed through CR and Cronbach's alpha. All constructs in the study exhibit values above the minimum threshold of 0.70. These values indicate that the indicators associated with each construct possess a sufficient level of internal consistency and reliably measure the same underlying concept [48]. Collectively, these results show that the constructs maintain an acceptable degree of measurement stability and consistency, ensuring that the structural model relationships can be interpreted with confidence. Therefore, the measurement instruments used in this study are considered to have achieved the required level of internal reliability [49, 50].

### 5.2 Discriminative Validity

The third step in evaluating the measurement model involves assessing discriminant validity. This assessment is conducted using the Fornell–Larcker criterion and the HTMT. A construct is considered to demonstrate adequate discriminant validity when its HTMT value is below 0.85 [49]. The results of this assessment are presented in Table 2.

Referring to Table 2, indicates that the HTMT values for all construct pairs fall below the threshold of 0.85. This demonstrates that the constructs are well differentiated and that each captures a concept distinct from the others [49]. From a conceptual standpoint, these findings verify that the constructs represent empirically separate dimensions, thereby supporting a valid interpretation of the relationships within the structural model. Consequently, the measurement model is considered to possess strong discriminant validity, enabling all latent constructs to proceed to the subsequent phase of analysis [50]. Thereafter, discriminant validity was further assessed using the Fornell–Larcker criterion. The results of this assessment are presented in Table 3.

Referring to Table 3, the Fornell–Larcker criterion indicates that the square root of the AVE for each construct is greater than its correlations with other constructs. This demonstrates that each construct is better represented by its own indicators than by indicators originating from other constructs. Collectively, these results confirm that each latent variable maintains a clear and unique empirical identity, as the indicators load exclusively on their designated



constructs. These findings provide strong confirmation that the measurement model attains robust discriminant validity in accordance with PLS-SEM guidelines [48, 50].

**Table 1.** The constructs, items, and operationalization

| Construct                       |       | Manifest Variable | Outer Loading | Mean  | STDEV | VIF   |
|---------------------------------|-------|-------------------|---------------|-------|-------|-------|
| Port infrastructure performance |       |                   |               |       |       |       |
| Cronbach's alpha                | 0.936 | PP1               | 0.779         | 5.660 | 0.863 | 4.500 |
|                                 |       | PP2               | 0.801         | 4.700 | 1.591 | 4.470 |
|                                 |       | PP3               | 0.793         | 3.880 | 1.669 | 3.762 |
|                                 |       | PP4               | 0.767         | 5.700 | 0.831 | 4.913 |
|                                 |       | PP5               | 0.797         | 2.760 | 1.379 | 3.199 |
|                                 |       | PP6               | 0.768         | 4.380 | 1.112 | 2.812 |
|                                 |       | PP7               | 0.801         | 5.860 | 0.825 | 3.704 |
|                                 |       | PP8               | 0.734         | 6.200 | 0.632 | 3.145 |
|                                 |       | PP9               | 0.771         | 6.340 | 0.738 | 4.906 |
|                                 |       | PP10              | 0.751         | 5.720 | 0.749 | 3.594 |
|                                 |       | PP11              | 0.773         | 5.980 | 0.836 | 4.835 |
|                                 |       | PP12              | 0.779         | 6.040 | 0.937 | 4.268 |
|                                 |       | PP13              | 0.783         | 6.020 | 0.860 | 3.522 |
| Logistics performance           |       |                   |               |       |       |       |
| Cronbach's alpha                | 0.944 | LP1               | 0.865         | 5.320 | 0.947 | 3.475 |
|                                 |       | LP2               | 0.905         | 5.240 | 1.069 | 4.347 |
|                                 |       | LP3               | 0.930         | 5.320 | 1.085 | 4.186 |
|                                 |       | LP4               | 0.901         | 5.400 | 0.894 | 3.421 |
|                                 |       | LP5               | 0.901         | 5.480 | 0.985 | 4.848 |
|                                 |       | LP6               | 0.883         | 5.580 | 1.022 | 3.871 |
|                                 |       | LP7               | 0.852         | 5.520 | 0.985 | 4.373 |
|                                 |       | LP8               | 0.790         | 5.580 | 0.874 | 3.193 |
|                                 |       | LP9               | 0.768         | 5.540 | 1.062 | 4.197 |
|                                 |       | LP10              | 0.793         | 5.480 | 1.005 | 4.581 |
|                                 |       | LP11              | 0.731         | 5.260 | 0.955 | 4.444 |
|                                 |       | LP12              | 0.750         | 5.380 | 0.797 | 3.962 |
| Economics growth                |       |                   |               |       |       |       |
| Cronbach's alpha                | 0.902 | EG1               | 0.749         | 5.140 | 0.749 | 2.210 |
|                                 |       | EG2               | 0.788         | 5.620 | 0.998 | 2.745 |
|                                 |       | EG3               | 0.816         | 5.480 | 1.063 | 3.781 |
|                                 |       | EG4               | 0.763         | 5.380 | 0.690 | 2.139 |
|                                 |       | EG5               | 0.753         | 5.520 | 0.943 | 2.043 |
|                                 |       | EG6               | 0.810         | 4.860 | 1.265 | 3.122 |
|                                 |       | EG7               | 0.759         | 5.940 | 0.810 | 2.331 |
|                                 |       | EG8               | 0.725         | 4.840 | 1.027 | 1.961 |

Note: STDEV: standard deviation; VIF: variance inflation factor; PP: port performance; LP: logistics performance; EG: economic growth; CR: composite reliability; AVE: average variance extracted.

### 5.3 Structural Model (Inner Model)

The analysis used to evaluate the structural model (inner model) involved collinearity analysis (VIF), path coefficients,  $R^2$ , and  $f^2$ . The first stage of evaluating the measurement model focuses on the coefficient of determination, or  $R^2$ , which quantifies the proportion of variance in the dependent variable explained by the independent variables in the model. For reflective latent variables,  $R^2$  serves as an indicator of the model's predictive power based on its indicators. According to Hair et al. [51],  $R^2$  values are classified as substantial when equal to or greater than 0.75, moderate when between 0.50 and 0.74, and weak when ranging from 0.25 to 0.49. The corresponding results are presented in Table 4.

The evaluation results indicate that port infrastructure performance and logistics performance collectively explain 57.4% of the variation in economic growth in Gorontalo Province. This value falls within the moderate to substantial category, indicating that Anggrek Port and its associated logistics system play a significant role in enhancing

regional economic activity. Port infrastructure contributes to the smooth flow of goods, reduces transportation costs, improves distribution efficiency, and expands market access. Nevertheless, 42.6% of the variation in economic growth is attributable to external factors. Meanwhile, the  $R^2$  value for logistics performance is only 7.6%, indicating that most of its variation is driven by factors outside the model. This finding suggests that logistics performance, as a dependent or primary mediating variable, is a multidimensional construct. In other words, port infrastructure performance alone is insufficient to fully account for its variability, indicating the presence of model misspecification in this study.

**Table 2.** Heterotrait–Monotrait Ratio (HTMT)

| Construct                       | Economics Growth | Logistics Performance | Port Infrastructure Performance |
|---------------------------------|------------------|-----------------------|---------------------------------|
| Economics growth                | –                | –                     | –                               |
| Logistics performance           | 0.396            | –                     | –                               |
| Port infrastructure performance | 0.762            | 0.204                 | –                               |

Note: Endashes indicate not recorded/no data.

**Table 3.** Fornell-Larcker criterion

|                                 | Economics Growth | Logistics Performance | Port Infrastructure Performance |
|---------------------------------|------------------|-----------------------|---------------------------------|
| Economics growth                | 0.771            | –                     | –                               |
| Logistics performance           | 0.407            | 0.842                 | –                               |
| Port infrastructure performance | 0.732            | 0.192                 | 0.777                           |

Note: Endashes indicate not recorded/no data.

**Table 4.**  $R$ -Square ( $R^2$ )

| Construct             | $R^2$ | Description |
|-----------------------|-------|-------------|
| Economics growth      | 0.574 | Moderate    |
| Logistics performance | 0.076 | Weak        |

#### 5.4 Path Coefficient

The assessment of path coefficients represents the final stage in the evaluation of the structural model. This stage, commonly referred to as hypothesis testing, examines whether the previously formulated hypotheses are empirically supported. Its primary objective is to determine whether the exogenous latent variables exert a statistically significant influence on the endogenous latent variables. Following the criteria proposed by Henseler et al. [49], a  $t$ -statistic greater than 1.96 indicates significance at the 95% confidence level ( $\alpha = 0.05$ ), and a  $p$ -value below 0.05 confirms a significant effect (hypothesis accepted). Conversely, a  $p$ -value greater than 0.05 indicates a non-significant effect (hypothesis rejected). The results of the hypothesis testing are presented in Table 5.

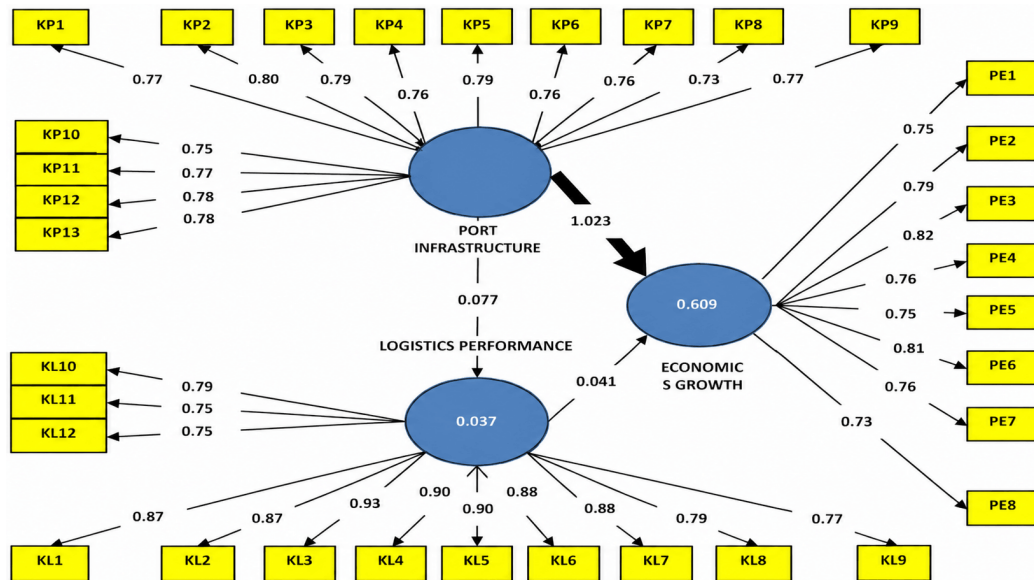
The interpretation of the path coefficients in Table 5 reveals several important findings. The analysis shows that port infrastructure performance exerts a strong and statistically significant positive effect on economic growth in Gorontalo Province. The path coefficient of 0.679 indicates that improvements in port infrastructure directly yield substantial gains in economic performance. This relationship is further reinforced by an  $R^2$  value of 0.574 and a very large effect size ( $f^2 = 1.023$ ), demonstrating that the infrastructure of Anggrek Port constitutes a critical component of the region's economic framework. Accordingly, Hypothesis 1 is supported. However, the effect of port infrastructure performance on logistics performance is not statistically significant. Although the positive coefficient of 0.236 suggests that infrastructure improvements may enhance logistics operations, the evidence does not indicate a meaningful impact. Thus, Hypothesis 2 is rejected. Similarly, logistics performance does not significantly influence economic growth in Gorontalo Province. Despite the positive coefficient, the absence of statistical significance, combined with the small effect size ( $f^2 = 0.041$ ) and low  $R^2$ , indicates that logistics plays only a marginal role,

functioning primarily as a supporting element rather than a key driver of economic growth. Therefore, Hypothesis 3 is rejected. The results of the significance testing are presented in Figure 2.

**Table 5.** Path coefficient

| Hypothesis (H) | Path                                                    | $\beta$ | $t$ -Statistic ( O/STDEV ) | $f^2$ | $p$ -Value | Variance Inflation Factor (VIF) |
|----------------|---------------------------------------------------------|---------|----------------------------|-------|------------|---------------------------------|
| H1             | Port infrastructure performance → economic growth       | 0.679   | 9.680                      | 1.023 | 0.000      | 1.038                           |
| H2             | Port infrastructure performance → logistics performance | 0.236   | 1.275                      | 0.077 | 0.256      | 1.000                           |
| H3             | Logistics performance → economic growth                 | 0.192   | 0.944                      | 0.041 | 0.375      | 1.038                           |

Note: STDEV: standard deviation.



**Figure 2.** The path structural model

## 6 Discussion

The results indicate a positive and statistically significant relationship between the performance of Anggrek Port infrastructure and economic growth in Gorontalo Province, with a very large effect size. This finding confirms that the development of physical port infrastructure serves as a primary driver of regional economic activity by improving the smooth flow of goods, increasing trade intensity, and enhancing the mobility of leading commodities, particularly within an economic structure still dominated by the primary sector and commodity-based trade. In this context, improvements in port infrastructure are able to generate direct economic impacts without relying on the readiness of complex logistics systems.

In contrast, the effect of infrastructure performance on logistics performance is very weak, reflecting that physical upgrades to the port have not been accompanied by a transformation toward an integrated and reliable logistics system. This logistics capability gap indicates that infrastructure advancement does not automatically translate into improved logistics performance when non-physical aspects such as governance, supply chain integration, information systems, supporting facilities, and institutional capacity have not developed adequately. A similar condition also explains the weak influence of logistics performance on economic growth, as logistics in Gorontalo continues to operate in a limited, high-cost, and non-integrated manner. Consequently, the findings reflect the reality that regional logistics



has not yet reached a level of institutional and systemic maturity sufficient to function as a primary lever of economic growth.

These findings indicate that the weak role of logistics can be attributed to institutional conditions, limited hinterland connectivity, and inadequate multimodal integration in Gorontalo. Weak institutional capacity, characterized by low inter-agency coordination, suboptimal governance and service digitalization, as well as weak regulation and stakeholder collaboration, has resulted in improvements to physical port infrastructure not being accompanied by gains in logistics service efficiency, as explained within the institutional economics perspective. At the same time, insufficient hinterland road networks that are inadequate to support the movement of goods from production centers to Anggrek Port hinder the formation of efficient and competitive supply chains. Furthermore, the absence of multimodal transport integration causes logistics activities to rely on a single, relatively high-cost mode of transport, thereby preventing efficiency gains even when port infrastructure is upgraded. Such conditions are common among feeder ports in 3T (frontier, outermost, and underdeveloped) regions such as Gorontalo Province, making port infrastructure development the most realistic policy option to implement in the short term.

### **6.1 Comparative Analysis with Previous Studies**

The findings of this study differ from several previous studies, particularly regarding the observation that port infrastructure performance does not influence logistics performance, which subsequently results in the logistics variable showing no significant effect on economic growth. Prior studies such as Lakshmanan [7], Munim and Schramm [9], and Wilmsmeier and Hoffmann [28] reported a positive relationship between infrastructure performance and logistics performance. This divergence may be attributed to the specific research context of Anggrek Port, which is a feeder port in Gorontalo Province and the sole port examined in this study. In contrast, previous studies were conducted in multiple countries or regions and typically involved major hub ports, thereby enabling comparative analyses that produced positive and robust results due to the inclusion of ports with higher levels of infrastructure maturity and operational complexity.

### **6.2 Contribution and Recommendation**

Empirically, the study provides concrete evidence that feeder ports in developing regions exhibit relationship patterns that differ from those of hub ports. This is important because most theoretical logistics models are constructed based on conditions observed in major ports, whereas feeder ports such as Anggrek operate within environments characterized by limited network capacity, underdeveloped multimodal transport, and a strong reliance on primary commodities. The findings, therefore, reinforce the argument that institutional context and regional economic structures fundamentally influence how infrastructure improvements translate into economic benefits.

Institutional strengthening should be positioned as a top priority through enhanced coordination among port operators, local governments, industry actors, transport companies, and port authorities. The implementation of integrated governance mechanisms, such as a shared data-based Port Community System, constitutes a critical foundation for ensuring that logistics activities operate as a coherent system and are capable of translating infrastructure improvements into enhanced logistics service performance. Furthermore, strengthening connectivity must extend beyond port infrastructure to encompass hinterland distribution networks, including production access roads, local logistics nodes, and cargo consolidation facilities, in order to ensure smooth upstream–downstream flows. Integrating the port with industrial zones also represents a strategic measure to promote value-added activities, enabling logistics not only to support the movement of raw commodities but also to play a more substantial role in generating sustainable economic impacts.

## **7 Conclusions**

This study confirms that the relationship between port infrastructure, logistics performance, and economic growth is non-linear, with regional economic performance being more responsive to the strengthening of physical port foundations than to internal logistics mechanisms that have not yet reached maturity. Port development has been shown to generate broad economic impacts through improvements in structural conditions, such as distribution efficiency and service capacity, even before regional logistics systems operate optimally.

The policy implications indicate that, in the short term, optimizing the physical infrastructure of Anggrek Port should be prioritized as a key driver of regional economic growth, given its rapid and tangible effects. Strengthening berthing capacity, storage facilities, land accessibility, and basic digital systems represents a strategic approach that aligns with Gorontalo's characteristics as a region at an early stage of feeder port development. Nevertheless, the weak contribution of logistics underscores the need for medium- and long-term policies focused on institutional reform, improved governance, cross-agency collaboration, and the integration of transport information systems to translate infrastructure advancements into sustainable logistics efficiency.

Looking ahead, future research should expand the scope of analysis and sample size, adopt a comparative approach across ports, and incorporate additional variables such as institutional quality, hinterland connectivity,

and multimodal integration. Such an approach is expected to yield a more comprehensive and realistic model for explaining the role of ports and logistics in regional economic development, while also providing a stronger empirical foundation for evidence-based policymaking.

### Author Contributions

Conceptualization: R.Z.M., M.A., and W.P.H.; methodology: R.Z.M., M.A., and W.P.H.; software: R.Z.M.; formal analysis: R.Z.M.; investigation: R.Z.M. and W.P.H.; writing—original draft preparation: R.Z.M.; writing—review and editing: R.Z.M., M.A., and W.P.H.; visualization: R.Z.M.; supervision: M.A. and W.P.H.; administration: R.Z.M.; funding acquisition: R.Z.M. All authors have read and approved the final published version of the manuscript.

### Data Availability

The data used to support the research findings are available from the corresponding author upon request.

### Conflicts of Interest

The authors declare no conflicts of interest.

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